RFHA-YOLO: Dynamic Receptive Field and Adaptive Hybrid Attention for Small-Object Detection in Remote Sensing Images

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***Abstract*—Small object detection in remote sensing images remains challenging due to limited feature resolution and complex backgrounds. Conventional detectors, due to fixed receptive fields and uniform attention, struggle to capture small-target features and suffer from background clutter. To address these limitations, we propose RFHA-YOLO, a novel method designed to enhance small-object feature extraction. RFHA-YOLO integrates two key innovations. First, a dynamic kernel decomposition module with an embedded receptive field attention block (RFA-Block) is constructed within the backbone as an alternative to the conventional spatial pyramid pooling structure. The core of RFA-Block is the RFAConv layer, which dynamically adjusts receptive fields across large-kernel branches to enable fine-grained detail preservation for small targets while suppressing irrelevant background regions. Second, an adaptive hybrid attention transformer (AHAT) is introduced in the feature fusion stage, enabling the adaptive integration of channel attention, spatial attention, and window-based self-attention. By learning importance scores at each pyramid level, this approach dynamically balances accuracy and computational efficiency in feature fusion. Extensive experiments on the DIOR, USOD, and AI-TODV2 aerial benchmarks have demonstrated to show that RFHA-YOLO outperforms several state-of-the-art detectors under similar model complexity. It achieves mAP_{50} scores of 84.8%, 72.3%, and 78.1%, reflecting overall detection accuracy, and impressive mAP_S scores of 28.8%, 10.0%, and 33.8%, which specifically measure performance on small objects. These results confirm the effectiveness of RFHA-YOLO for small object detection in remote sensing images, as its integrated RFA-Block and AHAT synergistically enhance small object feature capture and representation refinement.**

***Index Terms*—Deep learning, hybrid attention mechanism, receptive field, remote sensing, small object detection.**

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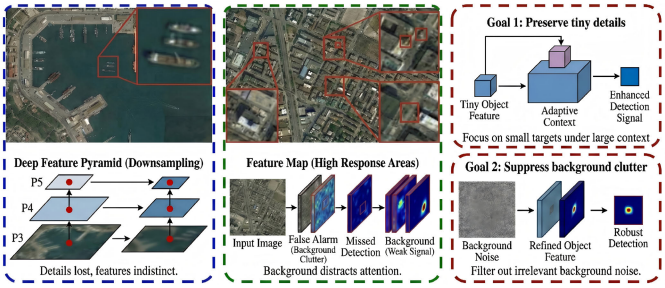
I. Introduction

Currently, small objects frequently appear in remote sensing images and are often essential targets for detection tasks. However, due to their inherently small size, low contrast, and insufficient information, small objects are highly susceptible to loss during downsampling, which poses significant challenges and leads to high miss rates in detection [1]. As a result, small object detection remains a central research topic in the remote sensing field, driven by its critical role in applications such as urban planning, maritime surveillance, traffic monitoring, and disaster response [2], [3], [4].

With the rise of high-resolution remote sensing images and the rapid advancement of deep learning techniques, object detection frameworks have undergone substantial transformation, evolving from traditional handcrafted approaches to highly optimized, data-driven convolutional architectures. Among these advancements, RCNN [5] successfully introduced convolutional neural networks (CNNs) to object detection, improving small object recognition accuracy. In addition, methods such as single shot multibox detector (SSD) [6], RetinaNet [7], and the detection transformer (DETR) seriesc [8], [9], [10], [11], [12] are proposed, incorporating multiscale feature pyramids, refined regression, and global modeling mechanisms to address challenges such as weak feature representation and object omission. In addition to the above frameworks, the you only look once (YOLO) series [13], [14], [15], [16], [17], [18] as a representative of single-stage detectors, has gained popularity in small object detection tasks due to its end-to-end structure and real-time inference capabilities. A representative, PP-YOLOE [19], leverages an anchor-free paradigm alongside an enhanced CSP-based backbone and an ELAN neck to strengthen feature extraction,

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(a) Challenge 1. (b) Challenge 2. (c) Desired capabilities.

Fig. 1. Illustration of key factors affecting small-object detection in remote sensing. (a) Tiny objects vanish after downsampling in deep feature layers. (b) Complex backgrounds cause confusion, leading to false alarms and missed detections. (c) Desired capabilities involve preserving tiny details and suppressing background clutter for robust detection.

thereby achieving high-precision detection for small objects while significantly reducing model parameters and inference overhead.

Despite the rapid progress of modern detectors, accurately identifying small objects in remote sensing images remains challenging due to two intrinsic factors, as summarized in Fig. 1. First, small objects often occupy only a few pixels (typically smaller than 32×32 [20]) and exhibit weak structural cues and low contrast, so their discriminative details are easily attenuated or even lost after repeated downsampling in deep feature pyramids [see Fig. 1(a)]. Second, remote sensing scenes are frequently dominated by cluttered backgrounds with textures similar to the targets, which distracts feature responses and leads to both false alarms and missed detections [see Fig. 1(b)]. Therefore, an effective small-object detector should simultaneously 1) preserve tiny details under large-context modeling and 2) suppress background clutter to produce robust detection signals, as illustrated in Fig. 1(c).

In response to these challenges, YOLO has been widely adopted for small object detection tasks owing to its fast inference speed and straightforward architecture that adapts well to dense prediction, prompting several studies to explore improvements within the YOLO framework. For instance, FFCA-YOLO [3] introduced a feature enhancement module (FEM), a feature fusion module (FFM), and a spatial context-aware module (SCAM) to enhance small object features and suppress background interference. SMN-YOLO [21] integrated coordinate attention and spatial-channel decoupled downsampling into a lightweight feature pyramid network, improving small object recognition. In addition, SOD-YOLOv10 [22] introduced a multidimensional interaction module for the Transformer backbone, enhancing global perception and integrating local and global information, improving detection performance for small targets.

Nevertheless, existing methods still face two fundamental limitations that hinder further performance improvements. On the one hand, the feature extraction process lacks effective modeling of information under varying spatial receptive fields, making it difficult to accurately represent the multiscale contextual regions where small objects reside. On the other hand, current feature fusion processes are typically based on static structures and lack mechanisms to selectively emphasize the most discriminative layers based on image content, resulting in insufficient expression of critical details and limited localization accuracy. Therefore, enhancing spatial detail perception and enabling selective modeling of important feature layers remain essential challenges in advancing small object detection for remote sensing images.

To address these challenges, we propose a small object method, RFHA-YOLO, an improved detector based on the PP-YOLOE architecture, tailored for small object detection in remote sensing images. We employ a dynamic receptive field method to adapt to complex background variations, and on this basis, introduce depthwise separable convolutions to enhance the spatial feature modeling capability, thereby improving the network's ability to process local context. In addition, we integrate an adaptive hybrid attention transformer (AHAT) in the neck of the PP-YOLOE network, which refines the feature representation using dual attention and window-based self-attention mechanisms. This module can adaptively select feature layers that contain the most detailed information about small objects, thereby enhancing feature resolution and detail restoration. Our contribution can be summarized as follows.

1) A previously underexplored contradiction between large receptive field expansion and fine-grained small-object detail preservation in spatial pyramid pooling is systematically characterized and theoretically formulated. This contradiction is alleviated by receptive-field-aware spatial weighting under large kernels, as instantiated by RFAConv in the final spatial pyramid pooling fast (SPPF) of PP-YOLOE, thereby reconciling contextual modeling with local detail retention.
2) A fundamental mismatch between static multiscale feature fusion and scale-dependent semantic importance in small-object detection is rigorously investigated and structurally reformulated. This mismatch is addressed via importance-aware attention allocation across feature pyramid levels, as realized by the AHAT strategy in the neck, enabling selective refinement of semantically critical features while avoiding unnecessary computational overhead.
3) Extensive experiments on the DIOR, USOD, and AI-TODV2 datasets demonstrate that receptive-field adaptivity and selective attention allocation constitute two complementary principles for robust small-object detection in complex remote sensing scenes. The effectiveness of these mechanisms is validated under comparable model complexity and inference efficiency.

II. Related Works

A. Small Object Detection

Small object detection in remote sensing images remains particularly challenging due to limited spatial resolution and complex backgrounds. One of the most significant challenges in small-object detection is the poor representation due to the limited information in small instances. A common approach to enhance detection performance is data augmentation via object oversampling, in which small objects of interest are duplicated [23]. This technique increased the likelihood of

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overlap between predicted and ground-truth labels, thereby improving precision. However, the overlap introduced during augmentation can degrade detection accuracy for other objects.

Recently, more studies have shifted from data-level augmentation toward feature-level enhancement and context-aware modeling for tiny objects. For SAR-based object detection, Li et al. [24] proposed space-frequency selection convolution to fully unleash channelwise potential, achieving strong representation under complex scattering environments. Wu et al. [25] integrated a Transformer into a dual-branch feature extraction design to jointly encode local details and global context, and further introduced a tiny-object detection head that reuses shallow high-resolution features to preserve fine-grained textures and positional cues. To further address multiscale representation challenges, Wu et al. [26] proposed M2FE-YOLO, which strengthened multiscale representations via a multibranch and multilevel feature enhancement strategy. This approach improved feature fusion effectively and reduced semantic inconsistencies across different pyramid levels, offering particular benefits for the detection of small objects in cluttered scenes. In addition, Zhou et al. [27] introduced a modality-aware disentangled feature modulation strategy, effectively strengthening shallow representations for dense tiny-object scenarios. Beyond remote sensing imagery, similar trends have been observed in dense crowd analysis. Dang et al. [28] designed multiobject behavior recognition frameworks based on advanced detection backbones, which effectively handle occlusions and scale variations. Their performance demonstrates the growing importance of structural feature enhancement instead of relying solely on data-level augmentation.

Compared to the aforementioned methods, enhancing small-object information and resolution through feature fusion combined with adaptive receptive-field modulation to suppress background interference yields superior performance.

B. Receptive Field for Context Enhancement

Fixed receptive fields employed in conventional CNNs struggle to adapt to varying scales and characteristics of remote sensing objects, especially small targets. To mitigate this limitation, several methods aim to adaptively modulate receptive fields. Dai et al. [29] introduced deformable convolutions, dynamically adjusting the spatial sampling locations within receptive fields based on the input features. Chen et al. [30] proposed dynamic convolution, which adaptively generates convolution kernels conditioned on the input, effectively improving feature extraction flexibility. Liu et al. [31] designed the receptive field block (RFB) in RFBNet, employing multibranch convolutions with varied kernel sizes to approximate the distribution of human visual receptive fields. Li et al. [32] proposed the selective kernel network (SKNet), which uses an attention mechanism to dynamically weight multiscale convolutional branches and thus adaptively select receptive field sizes. Ren et al. [33] proposed a hierarchical heterogeneous receptive-field design that explicitly emphasizes small-receptive-field features at shallow stages to strengthen local details for small objects, while suppressing large-receptive-field responses; the strategy is reversed in deeper layers to better capture large-object semantics. These approaches flexibly adjust convolutional kernel shapes and scales to preserve critical small-object details and enhance the multiscale responsiveness of detection networks.

While these traditional CNNs employ fixed-size receptive fields, making it challenging to differentiate scale variations between targets and backgrounds. While structures like the SPPF layer expand the receptive field, repeated pooling operations reduce spatial resolution, weakening small-object representation and context. A key challenge is designing a dynamic receptive-field-based pyramid pooling structure that adapts its pooling regions, enhancing the network's ability to preserve fine-grained object contours while mitigating background clutter and balancing receptive field expansion with spatial detail retention.

C. Feature Enhancement and Fusion Methods of Small Object Detection

Deep learning-based small object detection strongly depends on the quality of feature extraction and feature fusion. Due to continuous downsampling in modern backbones, the spatial resolution of deep feature maps becomes severely degraded, causing fine-grained structural cues of small targets to vanish. This motivates the development of advanced multiscale feature enhancement strategies that emphasize preserving discriminative details while integrating broader contextual information.

Recent research has focused on improving cross-scale representation quality beyond traditional adjacent-layer fusion. Yang et al. [34] proposed the asymptotic feature pyramid network (AFPN), which employs asymmetric convolution blocks to strengthen backbone features and enhance performance on objects with large aspect ratios. Zhu et al. [35] introduced DyFPN, where dynamic gating mechanisms adaptively select fusion pathways to achieve a more effective tradeoff between detection accuracy and computational efficiency. In addition, Fu et al. [36] explored hierarchical attention to learn level-specific feature contributions, enabling more discriminative fusion across scales.

Beyond architectural improvements, several recent works further enhance small-object features by integrating semantic priors and generative models into the multiscale fusion process. Zhang et al. [37] introduced semantic-aware mask guidance to preserve object contours and suppress background interference during fusion, demonstrating substantial gains on optical-SAR datasets. Meanwhile, Zhou et al. [38] incorporated diffusion-based structural priors into multiscale representations, effectively strengthening fine-grained details in cluttered SAR environments.

These studies highlight progress in adaptive, scale-aware feature enhancement for small object detection. However, the low resolution of deep feature maps still hinders the preservation of fine structural details, and existing methods remain limited in addressing large-scale variations. This motivates further exploration of more flexible and discriminative multiscale feature interaction strategies.

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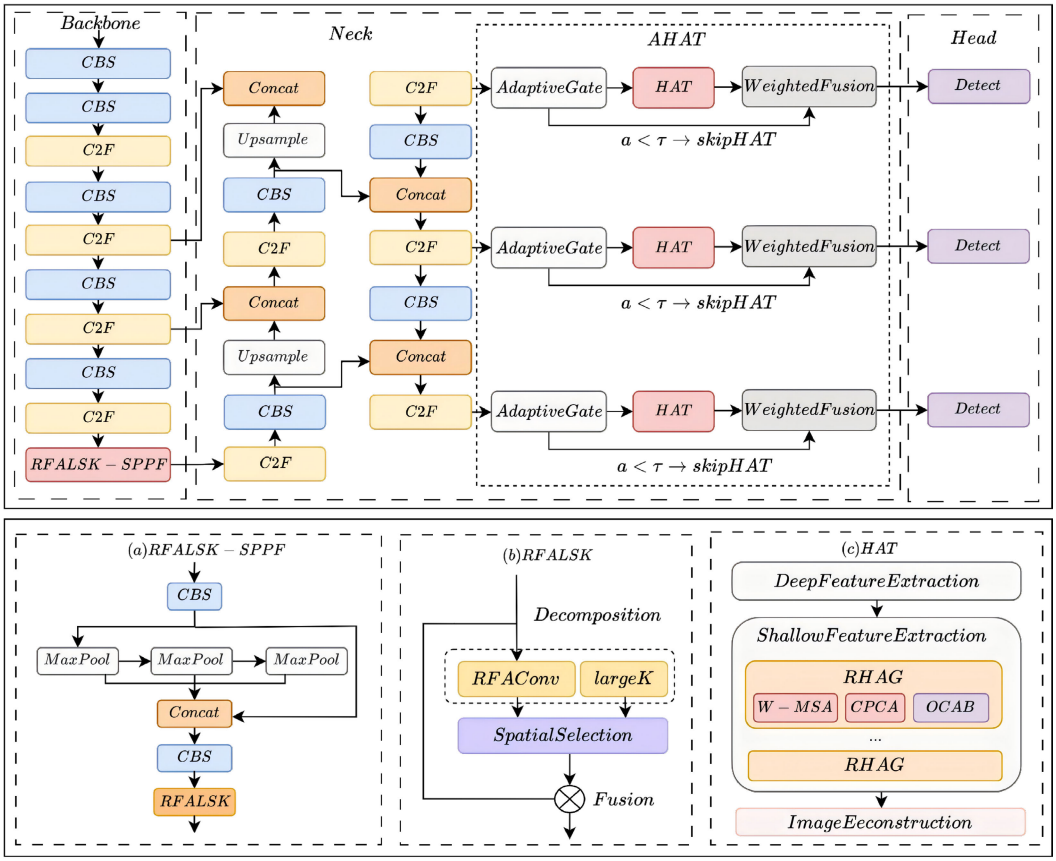


Fig. 2. Networkarchitecture of RFHA-YOLO. (a) RFALSK–SPPF. (b) RFALSK. (c) HAT.

III. Method

In this article, we propose a method, RFHA-YOLO, which is based on PPYOLOE [19]. The PPYOLOE framework ensures a favorable tradeoff between detection speed and accuracy in remote sensing images. As shown in Fig. 2, the proposed RFHA-YOLO architecture is composed of two major components. The first module introduces a dynamic kernel decomposition strategy, which integrates a dynamic receptive-field spatial pyramid pooling layer and a receptive field attention block (RFA-Block). This combination enables the network to adaptively adjust the receptive field to complex background conditions and preserve fine-grained spatial features. The second module incorporates an AHAT, which includes an enhanced HAT and an AHAT. This module selectively enhances semantic representations at different feature levels, allowing the network to focus on small-object details while maintaining computational efficiency.

A. Dynamic Receptive Field Module

To address the limitations of fixed receptive fields and the loss of spatial detail, we propose a novel dynamic receptive-field-based spatial pyramid pooling structure. Qualitative comparison between static and dynamic receptive fields is illustrated in Fig. 3(a). This structure is designed to adaptively adjust its pooling regions, which enhances the network's ability to retain fine-grained object contours and mitigate semantic confusion with background clutter. By effectively balancing receptive field expansion with the preservation of spatial resolution, our approach improves the detection of small objects in complex remote sensing scenes.

Building on the adaptive pooling framework, we employ large kernel attention, which provides a receptive field analogous to the self-attention mechanism. By utilizing depthwise convolution, dilated convolution, and 1×1 convolution, large convolutional kernels can be constructed efficiently with fewer parameters and reduced computations. Inspired by LSK-Net [39], we introduce the LSK-Block module after the SPPF (LSK-SPPF) layer to refine the context-enriched features produced by SPPF without further reducing spatial resolution, thereby compensating for the detail attenuation of tiny objects caused by pooling-based aggregation. The LSK-Block is structurally composed of two key components: a large-kernel spatial selection subblock and a feedforward subblock based on a multilayer perceptron (MLP). The spatial selection subblock employs a sequence of large-kernel convolutions alongside a spatial kernel selection mechanism, effectively enhancing the representation of fine-grained object contours and alleviating semantic confusion caused by complex backgrounds. Notably, the original LSK relies on fixed-weight large-kernel operators, which can be suboptimal for remote-sensing small targets with sparse and variable responses; such fixed kernels may blur weak target cues and be overwhelmed by background activations within large receptive fields. In particular, these large convolutional sequences are constructed through hierarchical decomposition into deep convolutions

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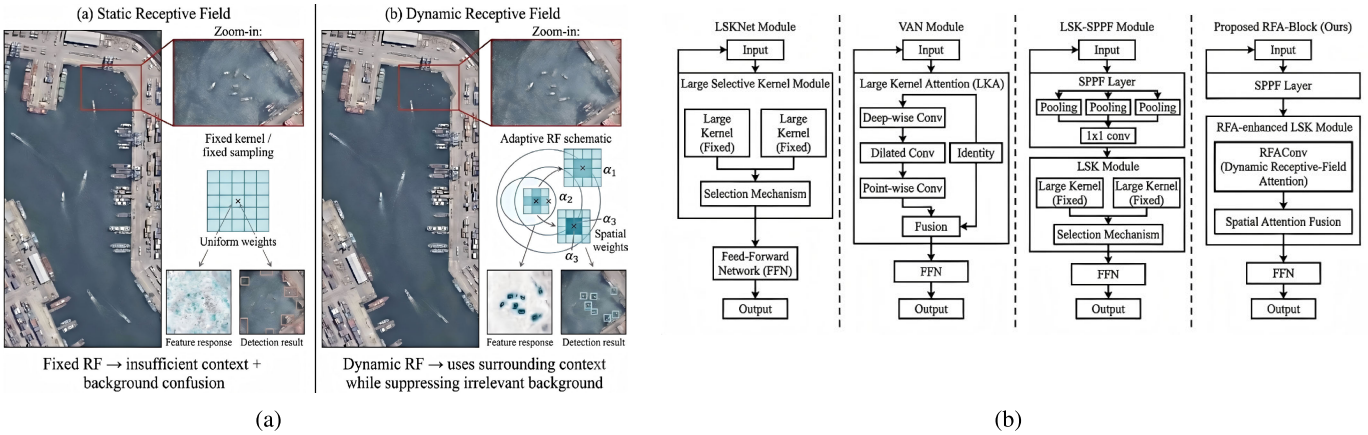


Fig. 3. Differentialcomparison of the methods. (a) Comparison of static and dynamic receptive fields. (b) Schematic comparison of large-kernel receptive-field modules. Unlike LSKNet, VAN, and LSK-SPPF, which use fixed kernels, the RFA-Block uses dynamic RFAConv to enhance small-object representation.

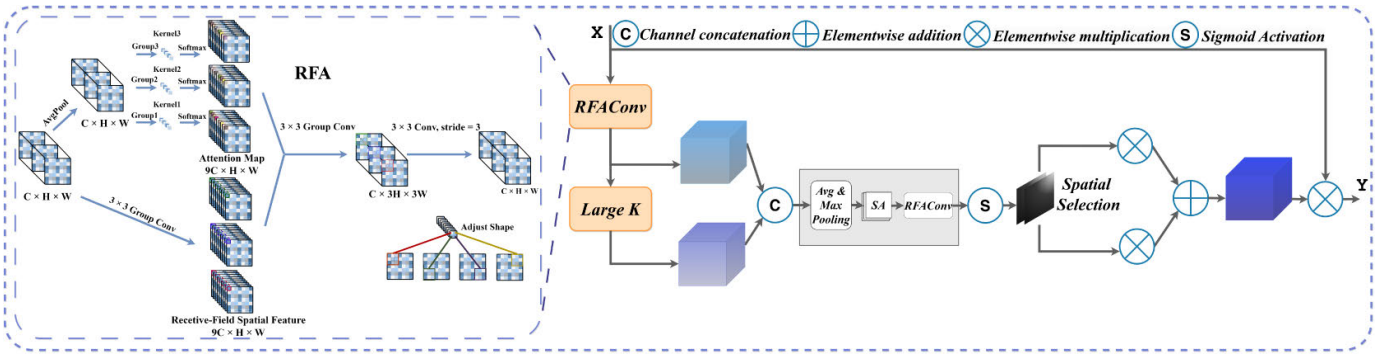


Fig. 4. Structure of the RFA-Block module.

with large kernel sizes and increasing dilation rates. We define the convolution kernel size as K, the expansion rate as d, and the receptive field RF as the area of the input image that influences the convolution operation. The initial expansion rate is set to 1, with an upper limit to prevent gaps between feature maps. Both the initial RF and the convolution kernel have the same initial size, and their relationship can be expressed as follows:

$$K_{i-1} \le K_i;\ d_1 = 1,\ d_{i-1} < d_i \le RF_{i-1} \tag{1}$$

$$RF_1 = K_1,\ RF_i = d_i\,(K_i - 1) + RF_{i-1}. \tag{2}$$

The formula indicates that the RF expands with an increase in both kernel size k and expansion rate d. The increasing kernel size and dilation rate ensure that the receptive field expands quickly enough. To capture the contextual information of the input feature X across different ranges, a series of deep convolutions with varying receptive fields is employed for decomposition. The decomposition nuclei are expressed as D_i ($D_0 = X$). Assuming that there are N decomposition nuclei, dw_i represents a deep convolution with kernel K_i and expansion rate d_i, denoted as C_i^{dw}. After decomposition, each kernel is processed by a 1×1 convolution layer $C_i^{1\times1}$ to generate a separate set of feature maps, denoted as $\tilde{D}_i$, which capture channelwise information. This decomposition process is illustrated in the following equation:

$$D_{i+1} = C_i^{dw}(D_i),\ \tilde{D}_i = C_i^{1\times1}(D_i),\quad i \in [1, N]. \tag{3}$$

While the LSK-SPPF module is designed to improve contextual representation for small objects, the original LSK relies on fixed kernel weights in the large-kernel pathway, which restricts its adaptability to diverse feature distributions and may blur weak tiny-object cues under complex backgrounds. To address this limitation, we introduce receptive field attention convolution (RFAConv) into the LSK-based spatial selection. In particular, we replace only the first large-kernel convolution with RFAConv, since this entry operator largely determines the subsequent spatial selection response. The receptive-field-aware dynamic weighting at the first stage recalibrates features before large-kernel aggregation, thereby suppressing background-dominated activations while preserving sparse and variable small-target responses. We retain the second large-kernel branch to preserve the original multidilation context modeling behavior and maintain architectural stability, while further dynamic replacement becomes less cost-effective given the already refined upstream features. The improved module is shown in Fig. 4. For a clearer comparison with existing designs, Fig. 3(b) summarizes representative large-kernel attention modules such as LSKNet, VAN, and LSK-SPPF. These methods primarily enhance contextual modeling through fixed large-kernel operators, which limit their flexibility in emphasizing small-object cues under complex backgrounds. In contrast, the proposed RFA-Block introduces receptive-field-aware dynamic weighting within the large-kernel pathway, thereby enhancing small-object representation while retaining contextual benefits.

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RFAConv enhances a standard 3×3 convolution by dynamically weighting each position within its receptive field. RFAConv begins by applying a grouped 3×3 convolution to extract local receptive-field features, generating an initial feature map. Next, average pooling is used to aggregate these spatial features into a compact context summary. A grouped 1×1 convolution then facilitates information exchange across channels, enriching the feature representation. Following this, a softmax operation generates an attention map over the 3×3 window, assigning higher weights to the most informative positions. Finally, the attention map is applied elementwise to the transformed receptive-field features, amplifying relevant patterns and suppressing irrelevant ones.

Building on this core receptive-field attention principle, the RFAConv framework incorporates two enhanced variants: RFCBAM and RFCA. These modified versions implement a $k \times k$ convolution with stride k instead of the original elementwise attention weighting, thereby expanding the contextual scope while reducing spatial redundancy. In particular, RFCBAM integrates a lightweight squeeze-and-excitation (SE) block in place of the standard channel-attention submodule, achieving additional reductions in both parameters and computational overhead. By shifting attention from coarse global weighting to fine-grained receptive-field modulation, both modules consistently improve performance across different network backbones.

B. Adaptive Feature-Enhanced Hybrid Attention Transformer

The HAT [40] was originally developed and validated in image super-resolution, demonstrating strong capability in feature refinement. In this work, we migrate its hybrid-attention formulation to object detection to enhance multiscale representations under cluttered remote-sensing backgrounds. However, applying HAT uniformly across all pyramid levels introduces nonnegligible computational overhead and is often unnecessary for semantically weak low-level features. To address this issue, we propose an AHAT, which improves the HAT building block with CPCA and selectively activates HAT across pyramid levels via a lightweight importance predictor, achieving a better accuracy–efficiency tradeoff. This levelwise, importance-guided activation is conceptually related to dynamic layer selection in feature pyramids (e.g., DyFPN), yet AHAT keeps the original fusion topology unchanged and instead performs conditional feature refinement by gating the execution strength of HAT at each level.

The original HAT integrates a channel attention block (CAB) and window-based self-attention to leverage local and global contextual cues. Nevertheless, CAB tends to assign spatially uniform channel importance, which may be suboptimal when small targets are sparse and background textures dominate. Therefore, we replace CAB with a *channel prior convolutional attention* (CPCA) block. CPCA employs multiscale depthwise separable convolutions to introduce spatially-aware channel modulation, enabling attention allocation jointly across channel and spatial dimensions. In particular, given an input feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, the modified hybrid attention block (HAB) is formulated as

$$\mathbf{Z} = \mathbf{X} + \mathrm{CPCA}\left(\mathrm{LN}\left(\mathbf{X}\right)\right) + \mathrm{W-MSA}\left(\mathrm{LN}\left(\mathbf{X}\right)\right) \tag{4}$$

$$\mathbf{Y} = \mathbf{Z} + \mathrm{FFN}\left(\mathrm{LN}\left(\mathbf{Z}\right)\right). \tag{5}$$

where LN(·) denotes layer normalization, W-MSA(·) represents (shifted) window-based multihead self-attention, CPCA(·) denotes channel prior convolutional attention, and FFN(·) is the feed-forward network.

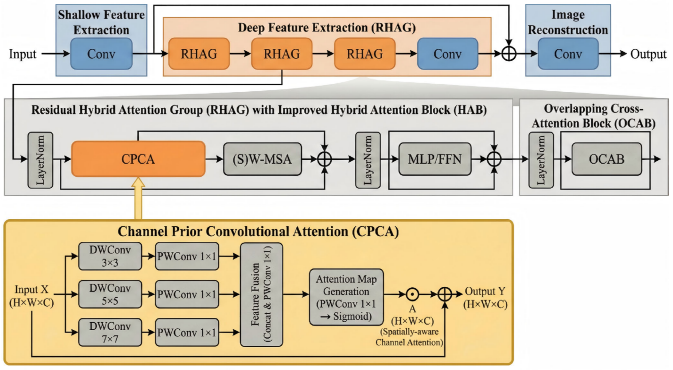


Fig. 5. Structure of the improved HAT module.

Meanwhile, we retain the overlapping cross-attention block (OCAB) from the original design to facilitate cross-window information exchange and mitigate boundary discontinuities caused by window partitioning. Following the standard HAT design, the module includes a shallow 3×3 convolution for detail-preserving feature projection and several residual hybrid attention groups (RHAGs) for deep refinement, as illustrated in Fig. 5. Importantly, our modification is strictly confined to the HAB, where the original CAB is explicitly replaced by CPCA as formulated above, while the OCAB pathway and overall RHAG topology remain unchanged.

After deep feature extraction, a global residual connection fuses shallow and deep features, ensuring that both low-level details and high-level semantics are retained. Finally, a lightweight reconstruction module (a small convolutional block) outputs the enhanced multiscale features, which are then propagated to the subsequent neck layers for further processing in the detection pipeline. Introducing hybrid attention improves multiscale feature refinement under cluttered backgrounds. Within the same refinement framework, CPCA further enhances background suppression and feature discrimination in complex remote-sensing scenes.

Despite the above improvements to HAT, which provide robust representational capacity by integrating channel attention and window-based self-attention mechanisms. However, applying HAT uniformly across all feature pyramid levels significantly increases computational overhead and reduces inference efficiency. To address this issue, we propose an AHAT module that selectively activates HAT according to the specific properties of each pyramid level. In particular, low-level features, which contain detailed spatial information but limited semantic content, bypass HAT processing to maintain computational efficiency. In contrast, high-level features,

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enriched with stronger semantic information, utilize HAT to enhance discriminative capability. Formally, for each pyramid level l (e.g., $l \in \{3, 4, 5\}$ corresponding to P3, P4, and P5), we first aggregate global contextual information from the feature map $\mathbf{F}_l \in \mathbb{R}^{C\times H\times W}$ via global average pooling, yielding a channel descriptor $\mathbf{u}_l$

$$\mathbf{u}_l = \mathrm{GAP}(\mathbf{F}_l) \in \mathbb{R}^C \tag{6}$$

then we calculate a scalar score s_l by passing $\mathbf{u}_l$ through a two-layer MLP

$$s_l = \mathbf{w}_2^\top \mathrm{ReLU}(\mathbf{W}_1 \mathbf{u}_l + \mathbf{b}_1) + b_2 \tag{7}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d\times C}$, $\mathbf{b}_1 \in \mathbb{R}^d$, $\mathbf{w}_2 \in \mathbb{R}^d$, and $b_2 \in \mathbb{R}$ are learnable parameters of the MLP, and ReLU$(\cdot)$ denotes the rectified linear unit. Note that $\mathbf{w}_2^\top$ represents the transpose of vector $\mathbf{w}_2$. A sigmoid activation then produces an importance weight $\alpha_l \in (0, 1)$

$$\alpha_l = \sigma(s_l) \in (0, 1) \tag{8}$$

where $\sigma(\cdot)$ is the sigmoid function.

Given α_l, the AHAT module fuses HAT-enhanced features and the original feature map $\mathbf{F}_l$ as follows:

$$\mathbf{F}_l^{\mathrm{out}} = \alpha_l \mathrm{HAT}(\mathbf{F}_l) + (1 - \alpha_l)\, \mathbf{F}_l. \tag{9}$$

When $\alpha_l \approx 1$, the lth level relies primarily on HAT's dual-attention fusion; when $\alpha_l \approx 0$, it bypasses HAT and retains the original feature map. Intermediate values yield a weighted combination of HAT and identity.

To further reduce inference cost, we introduce a *hard gating* threshold τ to enable conditional execution of HAT across pyramid levels. In all experiments, τ is fixed to 0.5, which is selected on the validation set to provide a stable accuracy–efficiency tradeoff and is then kept unchanged for all test-time evaluations to ensure reproducibility. In particular, we define a binary decision variable $g_l \in \{0, 1\}$ to represent the gating state of the lth level

$$g_l = \mathbb{I}(\alpha_l \geq \tau) = \begin{cases} 1, & \alpha_l \geq 0.5 \\ 0, & \alpha_l < 0.5 \end{cases} \tag{10}$$

where α_l denotes the importance weight of pyramid level $\mathbf{F}_l$.

The runtime procedure of AHAT, integrated with this hard gating mechanism, consists of two sequential steps.

1) *Importance Prediction:* For each pyramid level $\mathbf{F}_l$, compute the intermediate feature $\mathbf{u}_l$ via (6) first, and then derive the importance weight α_l through (7) and (8).
2) *Adaptive Fusion With Hard Gating:* Compare the computed α_l against the threshold τ. If $\alpha_l < \tau$, HAT is skipped by setting $\mathbf{F}_l^{\mathrm{out}} = \mathbf{F}_l$ (direct bypass); otherwise, the feature fusion is performed following (9).

This adaptive strategy allows AHAT to skip HAT on semantically weaker pyramid levels and concentrate refinement efforts on more critical high-level features, thereby improving inference efficiency without sacrificing representation capacity.

Building on the above rule, AHAT induces a gating pattern $\mathbf{g} = (g_3, g_4, g_5)$ over the different levels of the feature pyramid, specifically the low-level (P3), mid-level (P4), and high-level (P5) representations. Each g_l is determined independently by comparing α_l with the threshold τ. In practice, semantically weaker low-level features (e.g., P3) are more likely to be skipped, whereas semantically richer high-level features (e.g., P5) are more likely to activate HAT, thereby prioritizing computation on critical levels while avoiding redundant refinement on less informative ones.

By adaptively opening or closing HAT at each feature pyramid level, AHAT effectively balances computational cost while focusing attention resources on the most critical layers. This design maintains inference efficiency while significantly enhancing multiscale feature representation, particularly for challenging scenarios involving complex backgrounds and small object detection. As a result, the approach achieves notable improvements in overall detection performance.

IV. Experimental Results

A. Experimental Dataset Description

To sufficiently validate the effectiveness and superiority of the proposed small object detection method, we conducted experiments on public aerial image datasets, including DIOR, USOD, and AI-TODV2. The proportions of small, medium, and large objects in the three training sets are shown in Supplementary Fig. A1.

B. Model Training and Evaluation Metrics

All experiments in this article are programmed in Python, based on the Pytorch1.10.1 and CUDA11.3 environment for data processing. The experiments utilize an Intel[1] Xeon[1] Silver 4410Y @ 2.00 GHz, paired with one NVIDIA GTX 6000 GPU featuring 256 GB of memory, and operate on the Ubuntu 22.04 operating system. Default hyperparameters are set as follows: initial learning rate of 0.01, weight decay of 0.0005. For the DIOR and USOD datasets, the detector was trained over 140 epochs, whereas for AI-TODv2, training lasted 150 epochs. The experiments use PP-YOLOE pretrained weights obtained from the official COCO training dataset.

Mean average precision (mAP) is used as the standard evaluation metric, which can be divided into mAP_{50}, mAP_{75}, $\mathrm{mAP}_{50:95}$, and so on, according to the different IOUs. Here, mAP50 and $\mathrm{mAP}_{50:95}$ are used as the main evaluation metrics. sPs is used as the evaluation metric. In addition to accuracy, we report model complexity and inference efficiency in terms of *Params* (M), FLOPs (GB), *Time* (ms), and FPS (frames/s). Here, *Params* (M) denotes the number of trainable parameters, and FLOPs (GB) represents the computational complexity of the model. *Time* (ms) denotes the average per-image end-to-end inference latency at 640×640 with a batch size of 1 under the hardware and software environment described above, and FPS is obtained as FPS = 1000/Time(ms); thus, a smaller *Time* corresponds to a higher FPS. Furthermore, we employ scale-specific AP evaluations in AI-TODV2 [45], including AP_{vt}, AP_t, AP_s, and AP_m, which correspond to very tiny, tiny, small, and medium-sized object evaluations, as defined in [46]. Given that IoU is sensitive to position offsets for small objects, especially within the challenging AI-TODV2 dataset, we utilize optimal localization recall precision (oLRP) [47] to

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TABLE I
COMPARISONS OF STATE-OF-THE-ART OBJECT DETECTION METHODS ON DIOR DATASET

Methods	Backbone	mAP_{50-95}	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l	Time(ms)	FLOPs(GB)	Params(M)	FPS(Frames/s)
Deformable DETR [8]	ResNet50	46.4	66.6	49.8	9.8	34.5	67.6	29.8	58.42	40.05	85.4
DAB-DETR [9]	ResNet50	49.5	74.8	52.9	10.3	38.7	71.4	34.3	65.34	43.17	74.4
QO-DETR [10]	ResNet50	51.3	72.3	55.6	13.4	39.9	72.6	42.5	96.85	45.24	61.2
DINO [11]	ResNet50	57.5	80.5	62.9	21.6	47.3	76.3	50.6	179.0	47.58	36.8
DDQ [12]	ResNet50	57.8	80.4	63.7	22.3	46.0	76.3	67.0	180.0	48.38	16.8
GFL [42]	ResNet101	53.1	75.0	57.6	16.5	42.7	71.6	53.3	176.0	51.29	62.1
CentripetalNet [43]	Hourglass104	54.2	74.5	58.4	19.1	44.0	73.0	236.8	1502.0	206	29.3
RTMDet-s [44]	CSPNeXt	57.5	80.9	63.4	20.9	48.1	75.5	394.9	14.75	8.86	46.2
DiffusionDet [45]	ResNet50	53.4	77.2	58.6	18.4	43.8	71.5	56.3	180.0	48.38	57.7
YOLOv5 [13]	CSPDarknet	51.0	76.4	56.3	19.4	43.4	68.4	315.0	13.52	7.07	68.6
YOLOx [17]	CSPDarknet	55.9	81.8	62.2	22.8	47.5	72.2	85.1	13.40	8.97	30.3
YOLOv6 [14]	CSPDarknet	58.7	82.3	64.7	21.7	48.0	76.4	47.3	21.90	17.20	46.1
YOLOv7 [15]	CSPDarknet	57.0	80.9	63.4	25.2	49.1	73.0	16.5	51.90	36.61	53.1
YOLOv8 [16]	CSPDarknet	55.5	76.8	60.1	20.1	46.2	73.9	9.6	4.08	3.02	84.2
FFCA-YOLO [21]	CSPDarknet	58.2	82.9	64.3	24.8	49.6	76.9	18.9	23.6	8.10	52.9
YOLOv12 [18]	CSPDarknet	59.1	83.4	65.1	25.6	50.4	77.3	17.6	24.8	9.20	56.8
PPYOLOE [19]	CSPRepResNet	60.9	83.9	67.6	26.6	51.5	77.5	28.6	22.88	24.09	35.0
RFHA-YOLO (Ours)	CSPRepResNet	**62.1**	**84.8**	**68.7**	**28.8**	**52.4**	**78.5**	44.3	53.89	24.13	41.2

ensure a more reliable assessment of detection performance. Note that mAP is an accuracy measure where higher values indicate better performance, while oLRP is an error measure, where lower values indicate better performance.

C. Comparison With Other State-of-the-Art Detectors

We extensively compare RFHA-YOLO with multiple recent detection frameworks, including transformer-based methods (e.g., DAB-DETR, DINO, and DDQ), diffusion-based methods (e.g., DiffusionDet), and widely-used YOLO-series models (YOLOv5 through YOLOv12 and PP-YOLOE). Experiments are conducted on three challenging remote sensing datasets: DIOR, USOD, and AI-TODv2, covering diverse scenarios such as densely clustered small objects, extreme scale variations, and complex imaging conditions. The results clearly demonstrate RFHA-YOLO's consistent superiority in small-object detection, manifesting robust improvements across various performance metrics.

On the DIOR dataset, RFHA-YOLO achieves remarkable improvement, especially in detecting densely clustered small objects. In particular, as shown in Supplementary Table B1 and Table I, RFHA-YOLO improves the small-object-specific metric mAP_s by 2.2% (from 26.6% to 28.8%) compared to PP-YOLOE. The category-specific improvements in challenging classes like chimney (+3.9%), ship (+0.9%), and windmill (+1.8%) further illustrate our model's effectiveness in localizing small objects amidst complex backgrounds. Under the standard COCO-style metrics, RFHA-YOLO reaches 62.1% mAP_{50-95} (+1.2%), while also demonstrating strong performance on mAP_m (52.4%) and mAP_l (78.5%). Importantly, our method maintains real-time processing at 41.2 FPS, achieving a superior balance between high accuracy and computational efficiency.

Evaluation on the USOD dataset further demonstrates the superiority of RFHA-YOLO. As shown in Supplementary Table B2, our model establishes a new state of the art with 34.9% mAP_{50-95} and a remarkable 90.7% mAP_{50}, consistently outperforming the latest YOLOv12 and FFCA-YOLO. Notably, RFHA-YOLO maintains a competitive edge across all object scales, particularly in mAP_m and mAP_s, proving its robust detection capability and superior localization accuracy in challenging remote sensing scenarios.

The AI-TODv2 dataset, specifically designed for evaluating oriented and extremely small-object detection, places a high premium on metrics such as mAP_{vt}, mAP_t, and mAP_s. As illustrated in Supplementary Table B3, RFHA-YOLO establishes a new performance benchmark by achieving 28.5% mAP_{50-95} and a remarkable 64.5% mAP_{50}. Notably, our method exhibits superior sensitivity to the finest object scales, delivering substantial gains in the most challenging categories: it reaches 14.2% in mAP_{vt} and 29.1% in mAP_t, outperforming previous state-of-the-art models by 0.8% and 2.3%, respectively. These results underscore the effectiveness of the RFHA-YOLO architecture in capturing discriminative features for objects with extremely limited pixel information.

Furthermore, RFHA-YOLO demonstrates superior localization performance as evidenced by the oLRP sub-metrics in Supplementary Table B3. Notably, our method achieves an $oLRP_{IoU}$ of 31.7%, outperforming YOLOv12 (32.2%) and YOLOv8 (33.2%), which indicates a significantly higher regression precision for oriented small objects. In addition, the $oLRP_{FN}$ is reduced to 13.1%, highlighting a more robust recall capability compared to YOLOv8 (13.5%) and other mainstream detectors. This comprehensive optimization across error dimensions illustrates that RFHA-YOLO maintains high-fidelity spatial alignment without compromising detection

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TABLE II
COMPARISON BETWEEN LSK-BLOCK AND RFA-BLOCK ON DIOR DATASET

Variant	mAP_s ↑	GFLOPs↑	Params (M)↓	FPS ↑
LSK-Block	26.9	34.6	36.7	31.7
RFA-Block (Ours)	**27.7**	**34.7**	36.8	**31.9**

TABLE III
ABLATION STUDY OF CPCA-ENHANCED HYBRID ATTENTION WITH LEVELWISE GATING (AHAT) ON DIOR DATASET

Variant	mAP_{50} ↑	mAP_s ↑	Params (M)↓	FPS ↑
Baseline (HAT)	83.9	26.6	24.09	35.0
HAT (CAB + OCAB)	84.3	27.6	24.09	35.0
HAT† (CPCA + OCAB)	84.6	28.3	24.12	34.8
AHAT (CPCA + OCAB + gating)	**84.8**	**28.8**	24.13	**41.2**

sensitivity, providing a more reliable solution for complex small-object detection in remote sensing.

D. Ablation Experiments

Ablation experiments on the DIOR dataset validate each core component of RFHA-YOLO following a "component-specific detail → multicomponent summary" logic. Standalone module performance is first quantified, then its synergistic effects are analyzed, with all hyperparameters fixed to ensure reliability.

1) RFA-Block Performance Comparison With LSK-Block: To assess the contribution of the RFA-Block to small-object detection, it is compared with the widely used LSK-Block under controlled conditions—replacing the LSK-Block with the RFA-Block while keeping the baseline architecture and training parameters constant to isolate its independent performance improvement. As summarized in Table II, the lightweight design results in stronger feature representations, increasing mAPs by 0.8% (27.7 versus 26.9) with negligible changes in computation (GFLOPs: 34.7 versus 34.6) and parameters (36.8 versus 36.7 M), while boosting inference speed by 0.2 FPS (31.9 versus 31.7), demonstrating its effectiveness in expanding the receptive field for small targets without additional computational cost.

2) AHAT Ablation of CPCA Enhancement and Levelwise Gating: A progressive ablation study is conducted on the AHAT to analyze its two core components: CPCA and levelwise gating. Using the original HAT as the baseline, components are added sequentially to quantify individual gains, as shown in Table III. The results demonstrate clear improvements: replacing the conventional CAB with CPCA increases mAPs by 0.7% (27.6%–28.3%), confirming its effectiveness in enhancing small-target discriminative features. Furthermore, incorporating the gating mechanism yields an additional 0.5% mAPs gain and a significant 5.4-FPS speedup, with only a 0.01 M increase in parameters. This indicates that the gating mechanism filters redundant features, accelerates propagation, and strengthens useful semantics.

3) Multicomponent Synergy Comprehensive Ablation: To verify the synergy between RFA-Block and AHAT, a comprehensive ablation study was conducted by progressively integrating these components into the baseline model. This approach systematically evaluates the individual and combined contributions of each component. In particular, the RFA-Block and AHAT were sequentially incorporated into the baseline architecture to quantify their performance impact. Table IV summarizes the results across three scenarios, highlighting key metric variations with module inclusion/exclusion, where ✓ indicates adoption and × denotes removal.

Integrating the RFA-Block into the baseline model enhances performance across all metrics, particularly in small-object detection. In particular, mAP50 – 95 improves from 60.9% (baseline) to 61.5%, and mAP_s increases by 0.8% points, from 26.6% to 27.4%. These results demonstrate that the RFA-Block effectively expands the network's receptive field, allowing it to capture richer context around small targets while suppressing background noise.

The addition of AHAT further amplifies these gains. mAP50 – 95 rises from 60.9% to 62.1%, mAP_s increases by 2.2% points (26.6%–28.8%), and AR_s improves from 56.5% to 57.8%. AHAT refines high-level features through selective channelwise and spatial self-attention, enhancing target representation and enabling precise localization of small or weak targets in complex remote-sensing scenes.

4) Branch-Level Ablation Study: To assess the role of the RFALSK module at the 20 × 20 (P5) level, branch-level ablations are conducted on DIOR (see Table V). Removing the P5 branch degrades performance, especially for small objects, whereas relying solely on P5 severely impairs detection accuracy, underscoring the necessity of multiscale fusion. Eliminating the RFA module from P5 yields only marginal gains. By contrast, the full model achieves the best results, improving mAP_{50} and mAP_s by 0.9% and 2.2% over the baseline. This demonstrates that performance gains stem not merely from adding a branch, but from the receptive-field-aware refinement introduced by RFALSK.

5) Feature Map Visualizations: To elucidate the mechanisms driving performance gains, heatmap visualization was used to analyze spatial attention distribution. As shown in Fig. 6, the baseline model exhibits scattered activations, failing to focus on targets in cluttered remote-sensing backgrounds. In contrast, successive integration of RFA-Block and AHAT yields concentrated high-response regions aligned with targets. This visual evidence corroborates the quantitative results in Table IV, validating that each module enhances object localization and recognition in complex remote-sensing scenes.

The feature map visualizations demonstrate the impact of RFA-Block and AHAT on detecting small objects. In the baseline model, attention to small objects (e.g., boats and low-rise buildings) is diffused, making it difficult to distinguish them from the background. However, RFA-Block enhances target activation intensity and suppresses background interference. The combination of RFA-Block and AHAT further refines feature representation, sharpening small-target contours and enhancing background clutter suppression. This improved representation boosts target discriminability and enables more accurate localization of small/weak targets.

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TABLE IV
ABLATION STUDIES ON DIOR DATASET

Methods	baseline	HAT	AHAT	LSK-Block	RFAConv	RFA-Block	mAP_{50-95}	mAP_{50}	mAP_{75}	mAP_s	mAP_m	mAP_l
Panel A: Effect of HAT/AHAT												
RFHA-YOLO(ours)	✓	×	×	–	–	–	60.9	83.9	67.6	26.6	51.5	77.5
	✓	✓	×	–	–	–	61.5	84.3	68.1	27.6	52.0	68.0
	✓	✓	✓	–	–	–	**62.1**	**84.8**	**68.7**	**28.8**	**52.4**	**57.8**
Panel B: Effect of LSKBlock and RFAConv												
RFHA-YOLO(ours)	✓	–	–	×	×	–	60.9	83.9	67.6	26.6	51.5	77.5
	✓	–	–	✓	×	–	61.2	84.1	67.8	26.9	51.6	77.5
	✓	–	–	✓	✓	–	**61.4**	**84.4**	**68.0**	**27.7**	**51.9**	**77.8**
Panel C: Effect of RFA-Block and AHAT												
RFHA-YOLO(ours)	✓	–	×	–	–	×	60.9	83.9	67.6	26.6	51.5	77.5
	✓	–	×	–	–	✓	61.5	84.5	68.0	27.4	52.5	78.1
	✓	–	✓	–	–	✓	**62.1**	**84.8**	**68.7**	**28.8**	**52.4**	**78.5**

TABLE V
BRANCH-LEVEL ABLATION STUDY ON THE ROLE OF THE 20×20 (P5) FEATURE MAP ON DIOR DATASET

Model Variant	P5	RFA	mAP_{50}	mAP_{50-95}	mAP_s	mAP_m	mAP_l
Baseline (PP-YOLOE)	✓	×	83.9	60.9	26.6	51.5	77.5
w/o P5 Branch	×	–	82.1	60.1	23.4	50.8	77.2
P5 Only	✓	×	74.5	59.5	18.9	49.2	76.8
P5 w/o RFA	✓	×	84.1	61.1	27.0	51.8	77.6
Frozen P5 Gradients	✓	✓	83.5	60.6	25.8	51.2	77.4
Perturbed P5 ($\sigma = 0.1$)	✓	✓	82.9	60.0	24.7	50.9	77.3
Ours	✓	✓	**84.8**	**62.1**	**28.8**	**52.4**	**78.5**

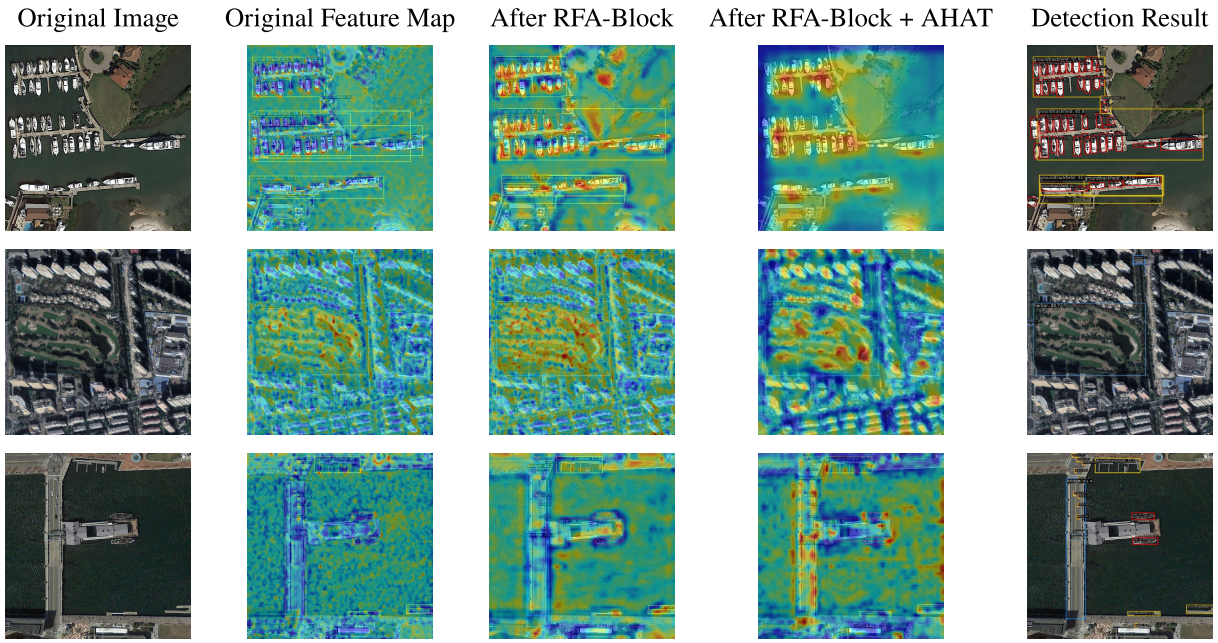


Fig. 6. Influence of RFA-Block and AHAT on feature extraction. The brighter color represents that the model pays more attention to that area.

E. Visualization of Detection Results

The qualitative results of RFHA-YOLO on three datasets (DIOR, USOD, and AI-TODv2) are shown in Fig. 7 and Fig. C1 (see Supplementary Material), together with representative baselines from the diffusion-series (DiffusionDet), YOLO-series (YOLOv12), and transformer-series (DDQ) detectors. For a clear visual comparison, detection outputs are overlaid as colored bounding boxes, and ROI zoom-ins are provided to highlight tiny objects. Overall, RFHA-YOLO yields more complete and stable detections, while competing methods exhibit noticeable false negatives and false positives in challenging scenes. On DIOR, our model maintains accurate localization across mixed object scales and complex

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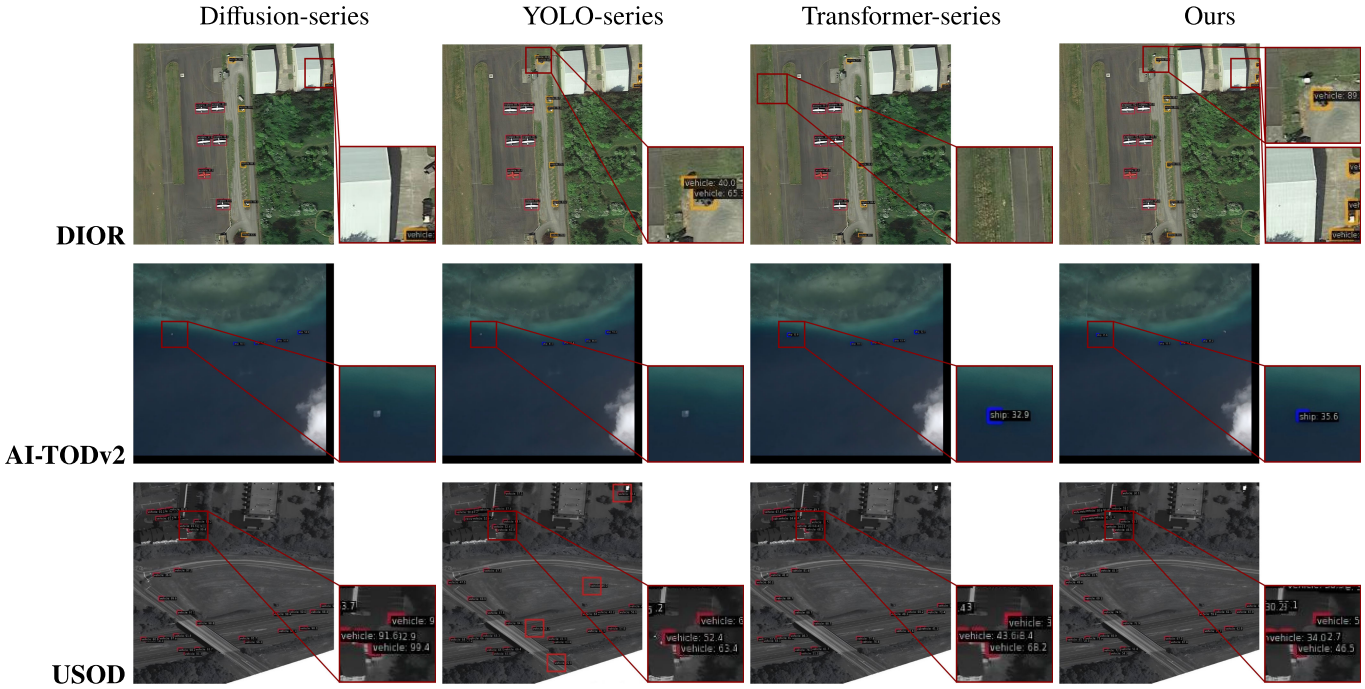


Fig. 7. Comparison of small-object detection results on remote sensing images.

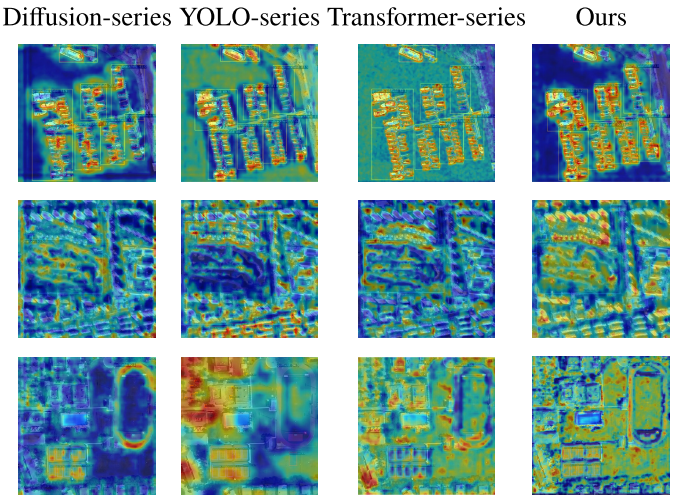


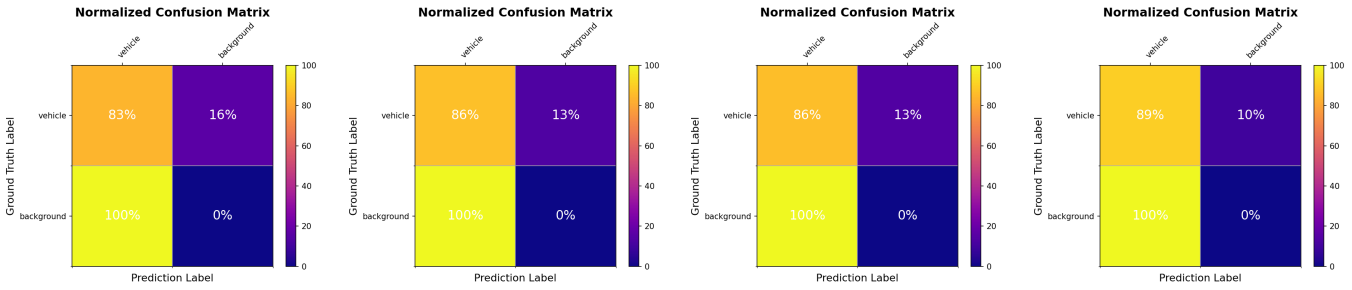
Fig. 8. Heatmap visualization comparison of small-object detection results on the DIOR dataset.

backgrounds, reducing missed small instances near cluttered structures. On AI-TODv2, RFHA-YOLO better separates densely packed tiny targets and preserves more valid detections in crowded regions, where other approaches tend to miss objects or produce fragmented/shifted boxes. On USOD, our method remains robust under low-light, shadows, and partial occlusions, suppressing background confusion and recovering small vehicles that are frequently missed by the compared detectors. These results demonstrate the superior small-object sensitivity and detection stability brought by our dynamic receptive-field modeling and adaptive hybrid attention.

To further evaluate the improvements brought by RFHA-YOLO on the DIOR dataset, we visualize the heatmaps from the final detection outputs of three model series, as shown in Fig. 8. The figure compares the heatmaps of Diffusion-series (DiffusionDet), YOLO-series (YOLOv12), and Transformer-series (DDQ). The results indicate that, in DIOR scenes, models such as DiffusionDet and YOLOv12 fail to consistently focus on discriminative target regions, leading to missed detections and false alarms, especially under complex backgrounds. In contrast, our method generates more compact and target-aligned activations with higher response concentration on salient regions, particularly for small-scale objects in DIOR. This improvement enhances decision reliability and boosts both detection accuracy and interpretability.

To comprehensively evaluate the performance of the proposed RFHA-YOLO model, we employ a confusion matrix for visual analysis. In this matrix, each row represents the ground-truth class distribution, while each column corresponds to the predicted class distribution by the model. The diagonal entries quantify the number of correct predictions where the predicted labels match the true labels, whereas the off-diagonal elements indicate misclassifications. As demonstrated in Fig. 9, the normalized confusion matrices compare four algorithm families: 1) the DiffusionDet series (represented by Diffusionet); 2) the YOLO series (represented by YOLOx); 3) the DETR series (represented by DDQ); and 4) the proposed method. Along the vehicle–vehicle diagonal, the proposed method attains an accuracy of 89%, whereas the DiffusionDet series achieves 83%. Concurrently, the vehicle-to-background misclassification rate decreases to 10% for the proposed method, compared to 16% for Diffusionet and approximately 13% for both YOLOx (YOLO series) and DDQ (DETR series). The lighter off-diagonal cells in subplot 4) indicate this reduced confusion between vehicle and background. Collectively, the confusion matrices show that the proposed model outperforms the three established families in vehicle detection, reducing misclassifications, particularly false negatives. The results also highlight significant improvements in distinguishing vehicles

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(a) (b) (c) (d)

Fig. 9. Confusion matrix of typical algorithms on USOD. (a) Diffusionet. (b) YOLOx. (c) DDQ. (d) Proposed Model.

from the background, with fewer vehicle-to-background misclassifications.

V. Conclusion

This article proposes RFHA-YOLO, an enhanced small-object detection framework for remote sensing images. A dynamic receptive-field module is integrated into the backbone to adaptively adjust pooling ranges, effectively preserving high-resolution details and improving localization accuracy for small targets amidst complex backgrounds. In addition, depthwise separable convolutions strengthen spatial feature modeling, enabling the network to capture essential local contextual information. An AHAT embedded in the network's neck combines dual-attention and window-based self-attention mechanisms to selectively refine multiscale features. AHAT dynamically allocates computational resources based on predicted importance weights, thus balancing inference efficiency and representational capability. Extensive experiments on DIOR, USOD, and AI-TODv2 datasets demonstrate that RFHA-YOLO substantially outperforms state-of-the-art methods, especially in dense, extremely small-object scenarios and challenging conditions like poor lighting or occlusion.

Future work will explore lightweight quantization and structured pruning techniques to adapt RFHA-YOLO for resource-constrained platforms, aiming to achieve an optimal tradeoff between model compactness and detection precision. In addition, unsupervised pretraining on large-scale unlabeled remote sensing data may further enhance the model's robustness to varying environmental conditions.

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